پوهنتون کاردان
KARDAN UNIVERSITY

MODERN PROGRAMMING LANGUAGE (Python)

Starting Programming in Python

Boolean Expressions and Relational Operators

- **<u>Boolean expression</u>: expression tested by if statement to determine if it is true or false**
  - Example: a > b
    - `true` if a is greater than b; `false` otherwise
- **<u>Relational operator</u>: determines whether a specific relationship exists between two values**
  - Example: greater than (>)

Boolean Expressions and Relational Operators (cont'd.)

- **>= and <= operators test more than one relationship**
  - It is enough for one of the relationships to exist for the expression to be true
- **== operator determines whether the two operands are equal to one another**
  - Do not confuse with assignment operator (=)
- **!= operator determines whether the two operands are not equal**

Boolean Expressions and Relational Operators (cont'd.)

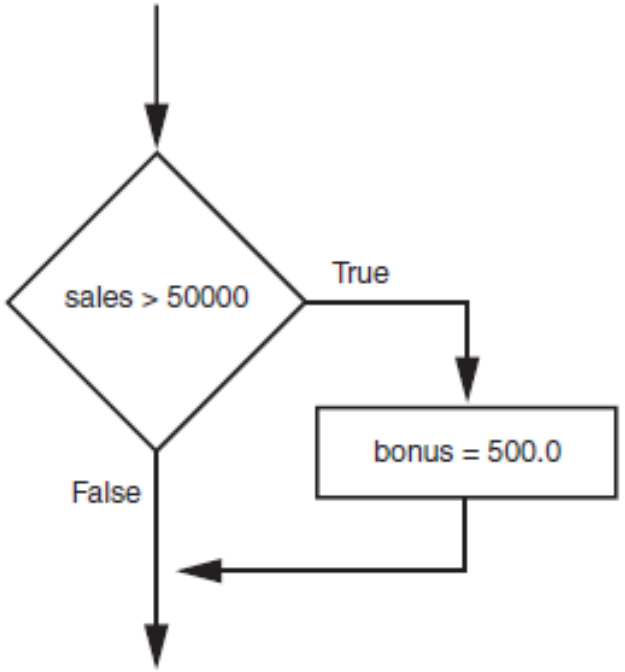
Table 4-2 Boolean expressions using relational operators

Expression	Meaning
`x > y`	Is x greater than y?
`x < y`	Is x less than y?
`x >= y`	Is x greater than or equal to y?
`x <= y`	Is x less than or equal to y?
`x == y`	Is x equal to y?
`x != y`	Is x not equal to y?

Boolean Expressions and Relational Operators (cont'd.)

- **Using a Boolean expression with the > relational operator**

Figure 4-3 Example decision structure



Boolean Expressions and Relational Operators (cont'd.)

- **Any relational operator can be used in a decision block**
  - Example: `if balance == 0`
  - Example: `if payment != balance`
- **It is possible to have a block inside another block**
  - Example: `if` statement inside a function
  - Statements in inner block must be indented with respect to the outer block

Bitwise operators

OPERATOR	NAME	DESCRIPTION	SYNTAX
&	Bitwise AND	Result bit 1,if both operand bits are 1;otherwise results bit 0.	x & y
|	Bitwise OR	Result bit 1,if any of the operand bit is 1; otherwise results bit 0.	x | y
~	Bitwise NOT	inverts individual bits	~x
^	Bitwise XOR	Results bit 1,if any of the operand bit is 1 but not both, otherwise results bit 0.	x ^ y
>>	Bitwise right shift	The left operand's value is moved toward right by the number of bits specified by the right operand.	x>>
<<	Bitwise left shift	The left operand's value is moved toward left by the number of bits specified by the right operand.	x<<

Identity Operators

- Identity operators compare the memory locations of two objects

Operator	Description	Example
is	Evaluates to true if the variables on either side of the operator point to the same object and false otherwise.	x is y, here is results in 1 if id(x) equals id(y).
is not	Evaluates to false if the variables on either side of the operator point to the same object and true otherwise.	x is not y, here is not results in 1 if id(x) is not equal to id(y).

Membership operators

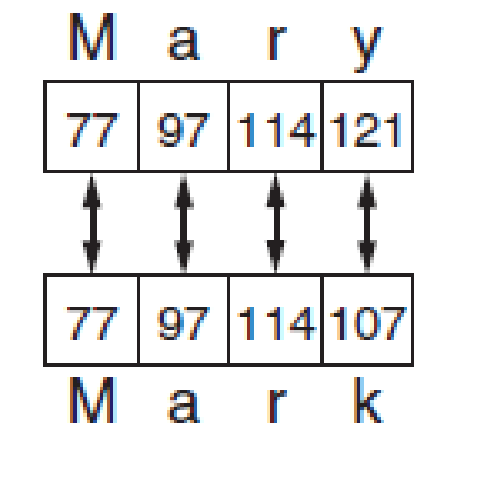
- Python offers two membership operators to check or validate the membership of a value.
- It tests for membership in a sequence, such as strings, lists, or tuples
- **in operator:** The ‘in’ operator is used to check if a character/ substring/ element exists in a sequence or not
- **‘not in’ operator-** Evaluates to true if it does not finds a variable in the specified sequence and false otherwise

Comparing Strings

- **Strings can be compared using the == and != operators**
- **String comparisons are case sensitive**
- **Strings can be compared using >, <, >=, and <=**
  - Compared character by character based on the ASCII values for each character
  - If shorter word is substring of longer word, longer word is greater than shorter word

Comparing Strings (cont'd.)

Figure 4-11 Comparing each character in a string



Logical Operators

- **<u>Logical operators</u>: operators that can be used to create complex Boolean expressions**
  - `and` operator and `or` operator: binary operators, connect two Boolean expressions into a compound Boolean expression
  - `not` operator: unary operator, reverses the truth of its Boolean operand

The `and` Operator

- **Takes two Boolean expressions as operands**
  - Creates compound Boolean expression that is true only when both sub expressions are true
  - Can be used to simplify nested decision structures
- **Truth table for the `and` operator**

Expression	Value of the Expression
false and false	false
false and true	false
true and false	false
true and true	true

The `or` Operator

- **Takes two Boolean expressions as operands**
  - Creates compound Boolean expression that is true when either of the sub expressions is true
  - Can be used to simplify nested decision structures
- **Truth table for the `or` operator**

Expression	Value of the Expression
false or false	false
false or true	true
True or false	true
true or true	true

Short-Circuit Evaluation

- **<u>Short circuit evaluation</u>: deciding the value of a compound Boolean expression after evaluating only one sub expression**
  - Performed by the `or` and `and` operators
    - For `or` operator: If left operand is true, compound expression is true. Otherwise, evaluate right operand
    - For `and` operator: If left operand is false, compound expression is false. Otherwise, evaluate right operand

The `not` Operator

- **Takes one Boolean expressions as operand and reverses its logical value**
  - Sometimes it may be necessary to place parentheses around an expression to clarify to what you are applying the not operator
- **Truth table for the `not` operator**

Expression	Value of the Expression
true	false
false	true

Checking Numeric Ranges with Logical Operators

- **To determine whether a numeric value is within a specific range of values, use `and`**
  - Example: `x >= 10 and x <= 20`
- **To determine whether a numeric value is outside of a specific range of values, use `or`**
  - Example: `x < 10 or x > 20`

Boolean Variables

- **<u>Boolean variable</u>: references one of two values, `True` or `False`**
  - Represented by `bool` data type
- **Commonly used as flags**
  - <u>Flag</u>: variable that signals when some condition exists in a program
    - Flag set to `False` → condition does not exist
    - Flag set to `True` → condition exists